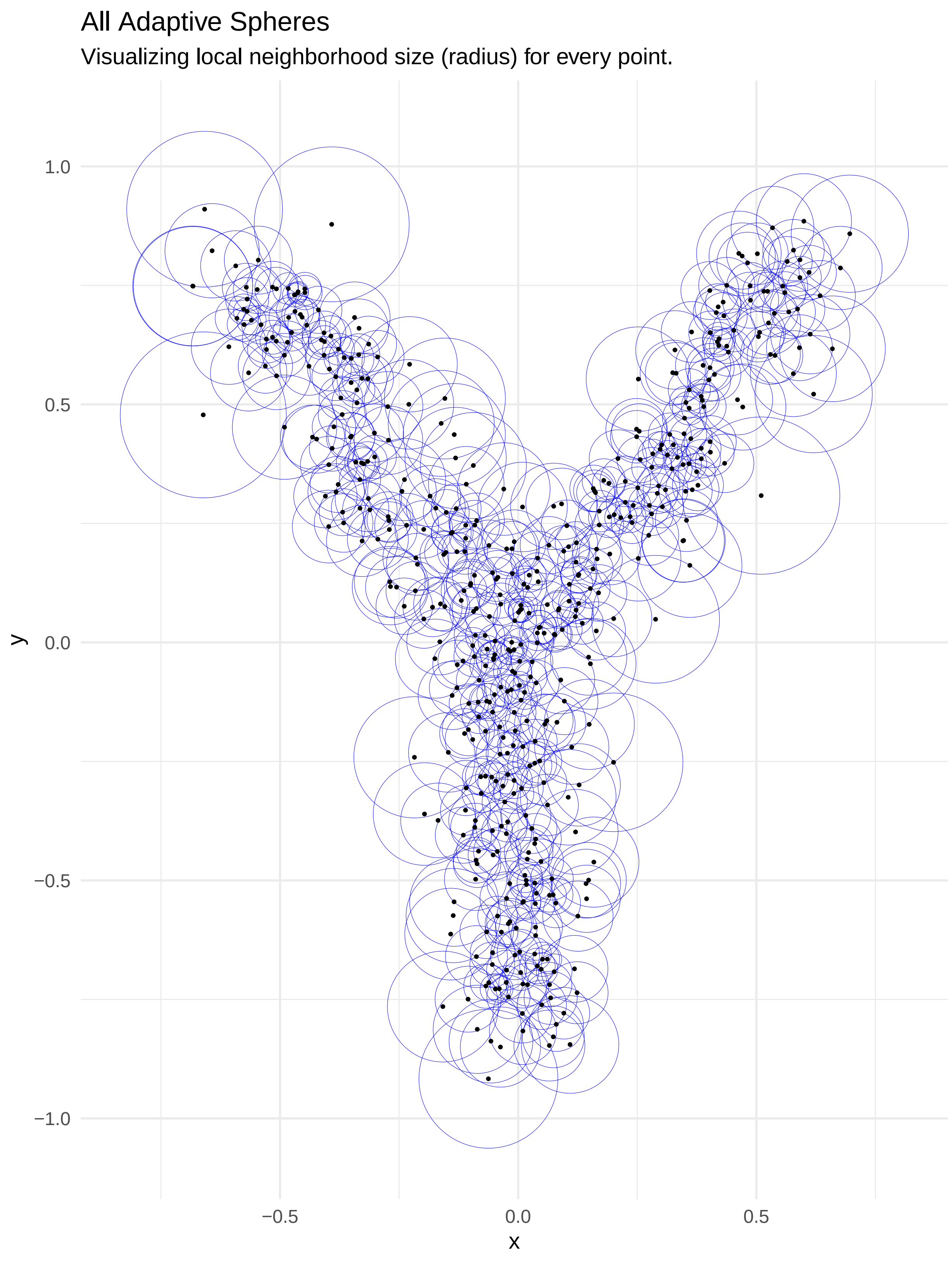
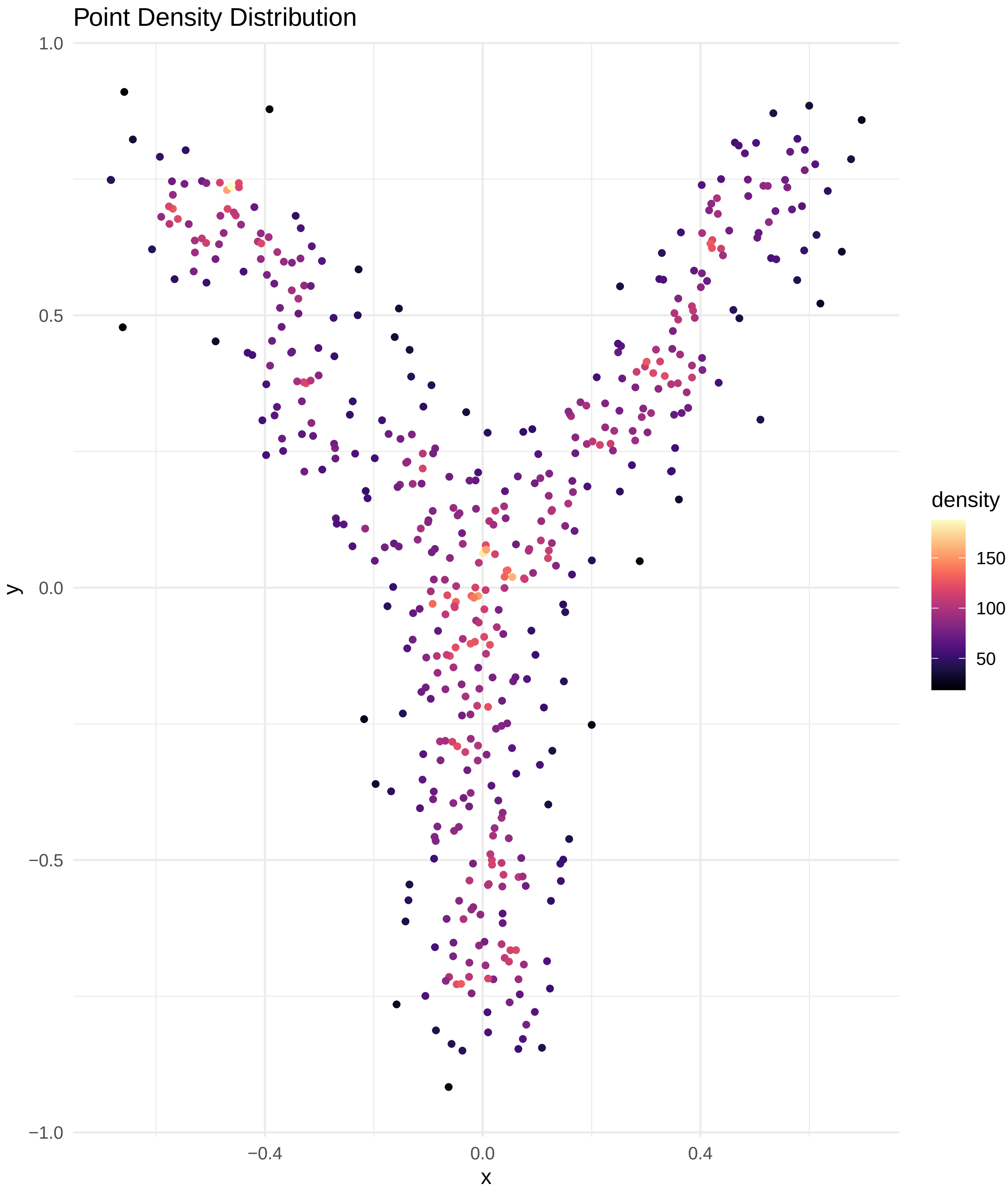
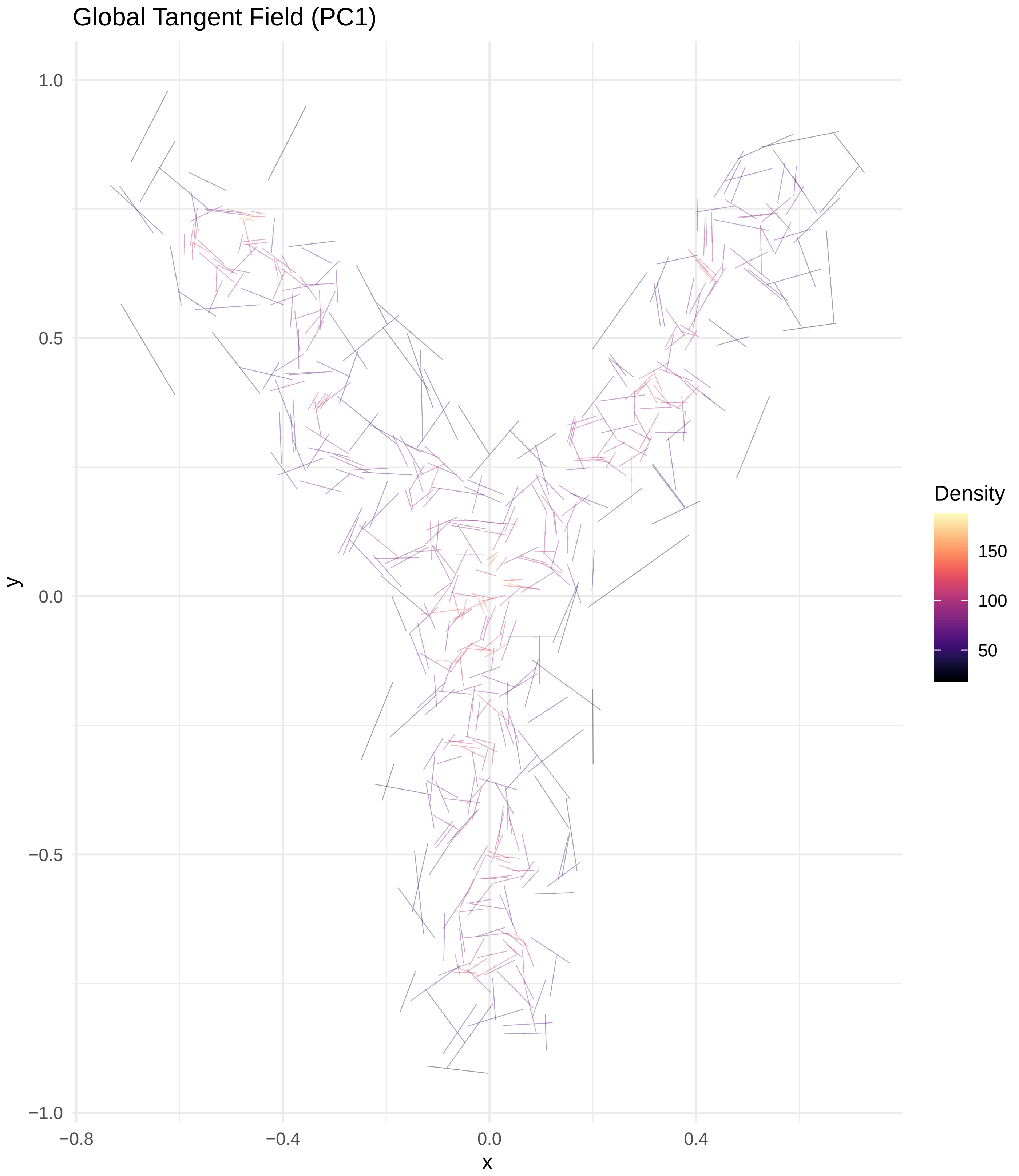
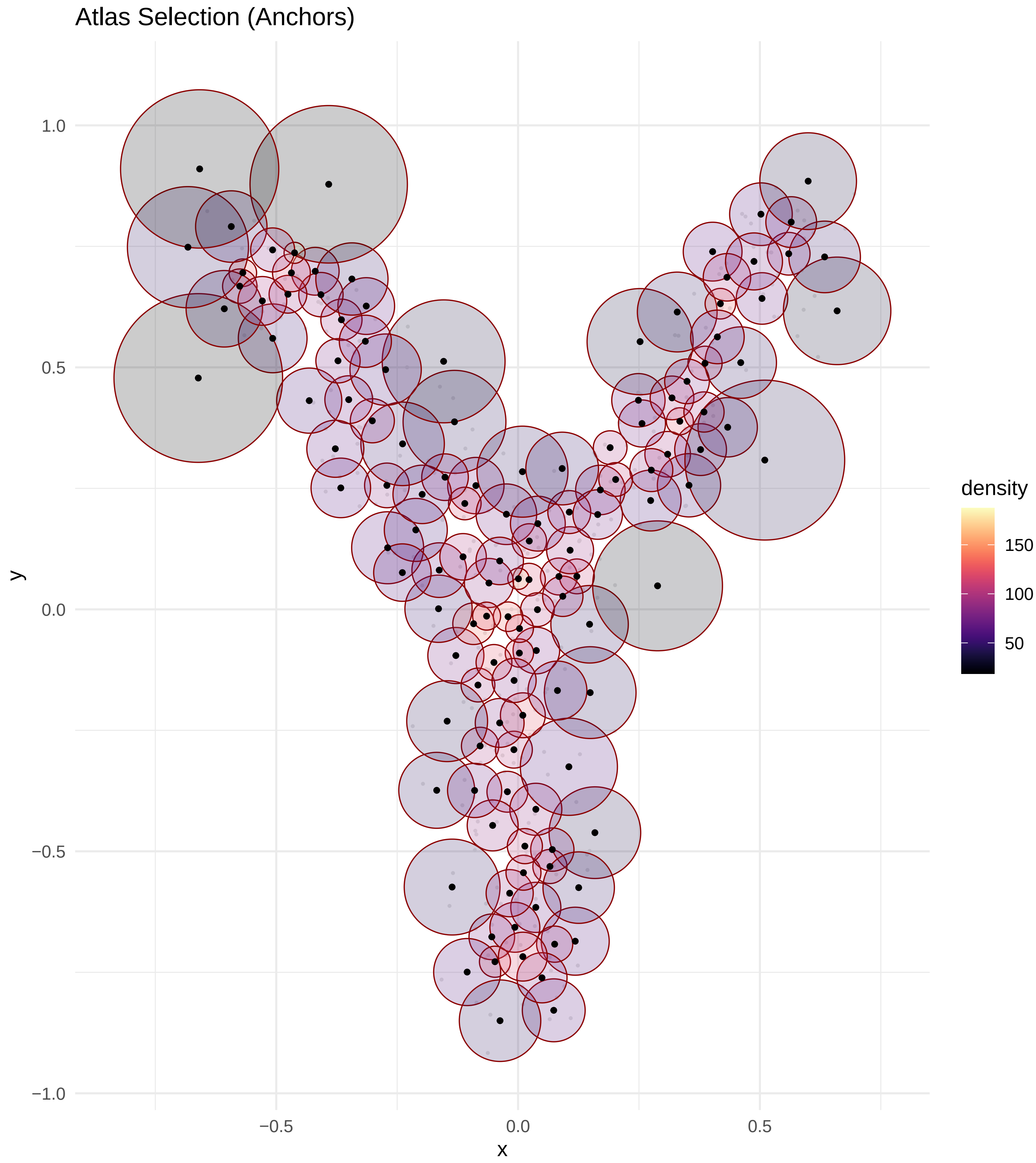

All Adaptive Spheres
Visualizing local neighborhood size (radius) for every point.
1.0
0.5
0.0
−0.5
−1.0
y
−0.5
0.0
0.5
x


Point Density Distribution
1.0
0.5
0.0
−0.5
−1.0
y
−0.4
0.0
0.4
x
density
150
100
50

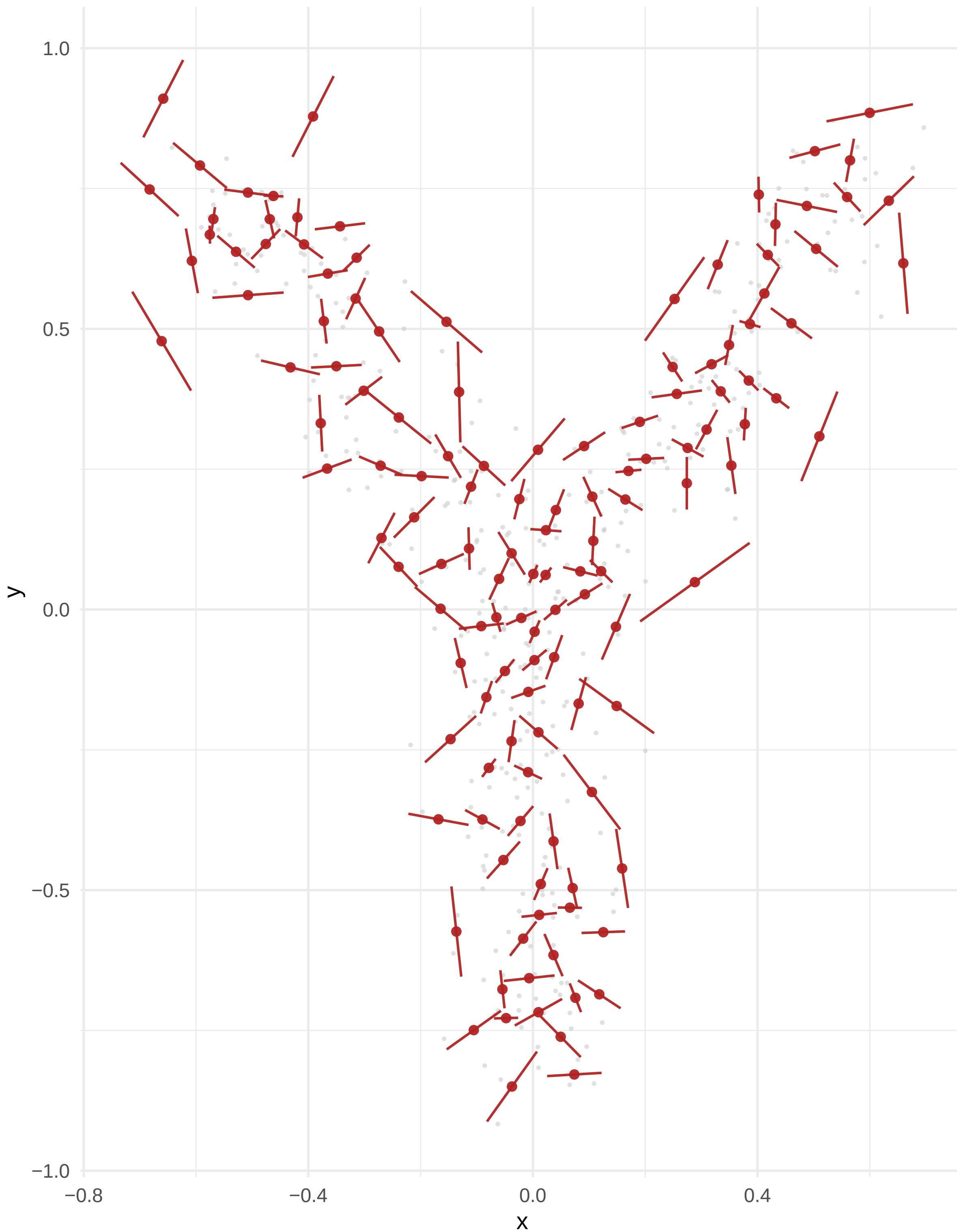

Global Tangent Field (PC1)
1.0
0.5
0.0
−0.5
−1.0
y
−0.8
−0.4
0.0
0.4
x
Density
150
100
50

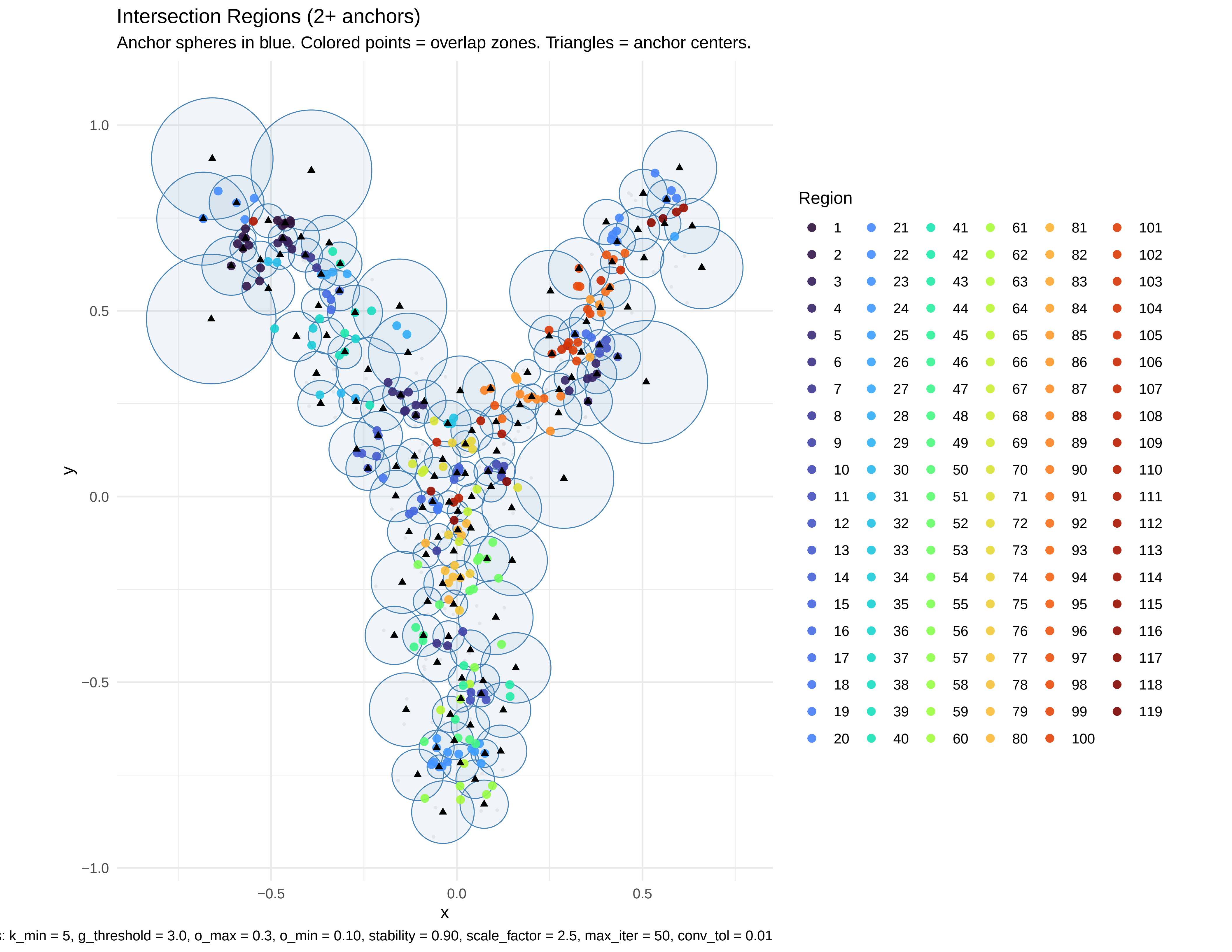


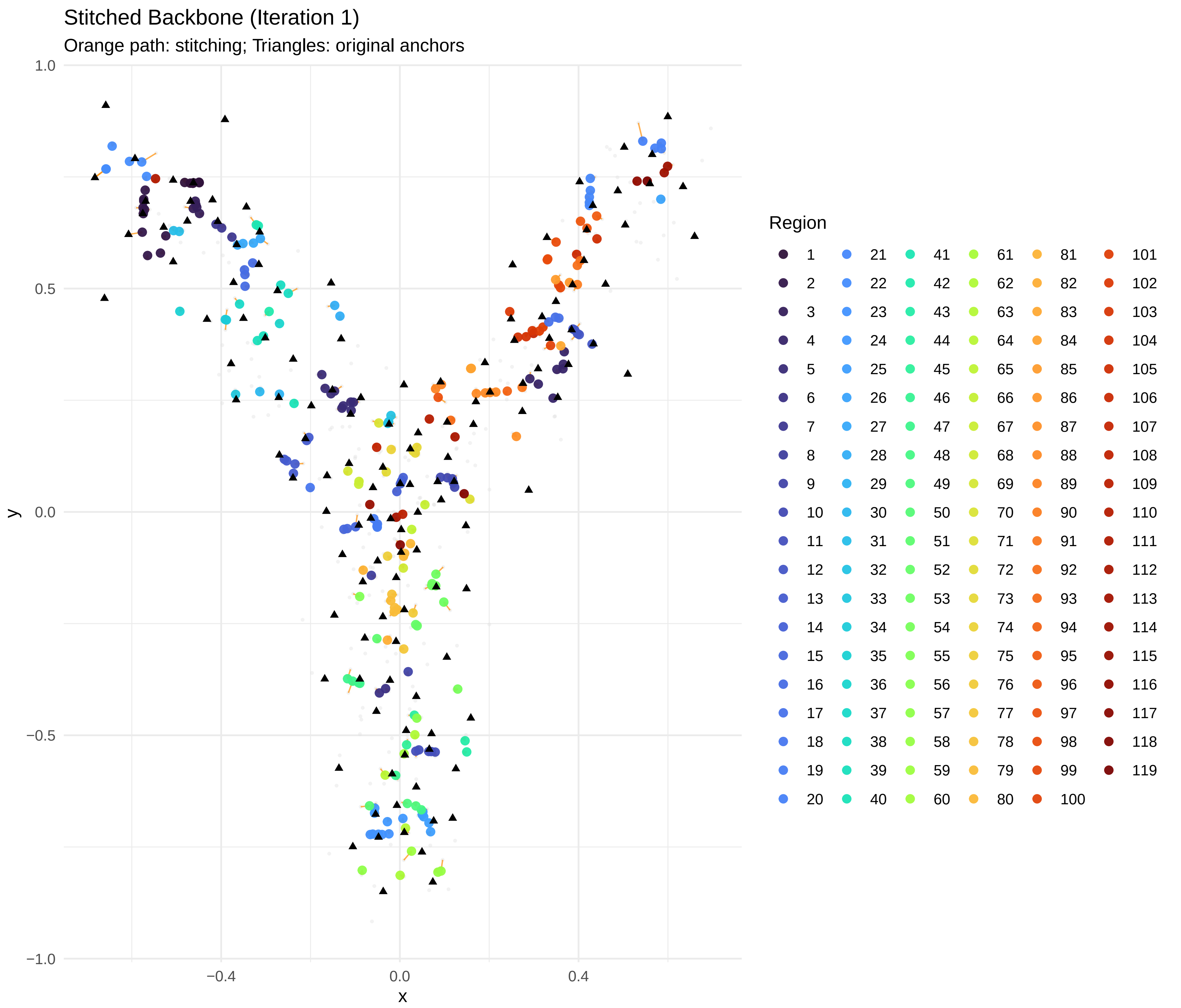
Atlas Selection (Anchors)
1.0
0.5
0.0
−0.5
−1.0
y
−0.5
0.0
0.5
x
density
150
100
50

Anchor SVD Segments

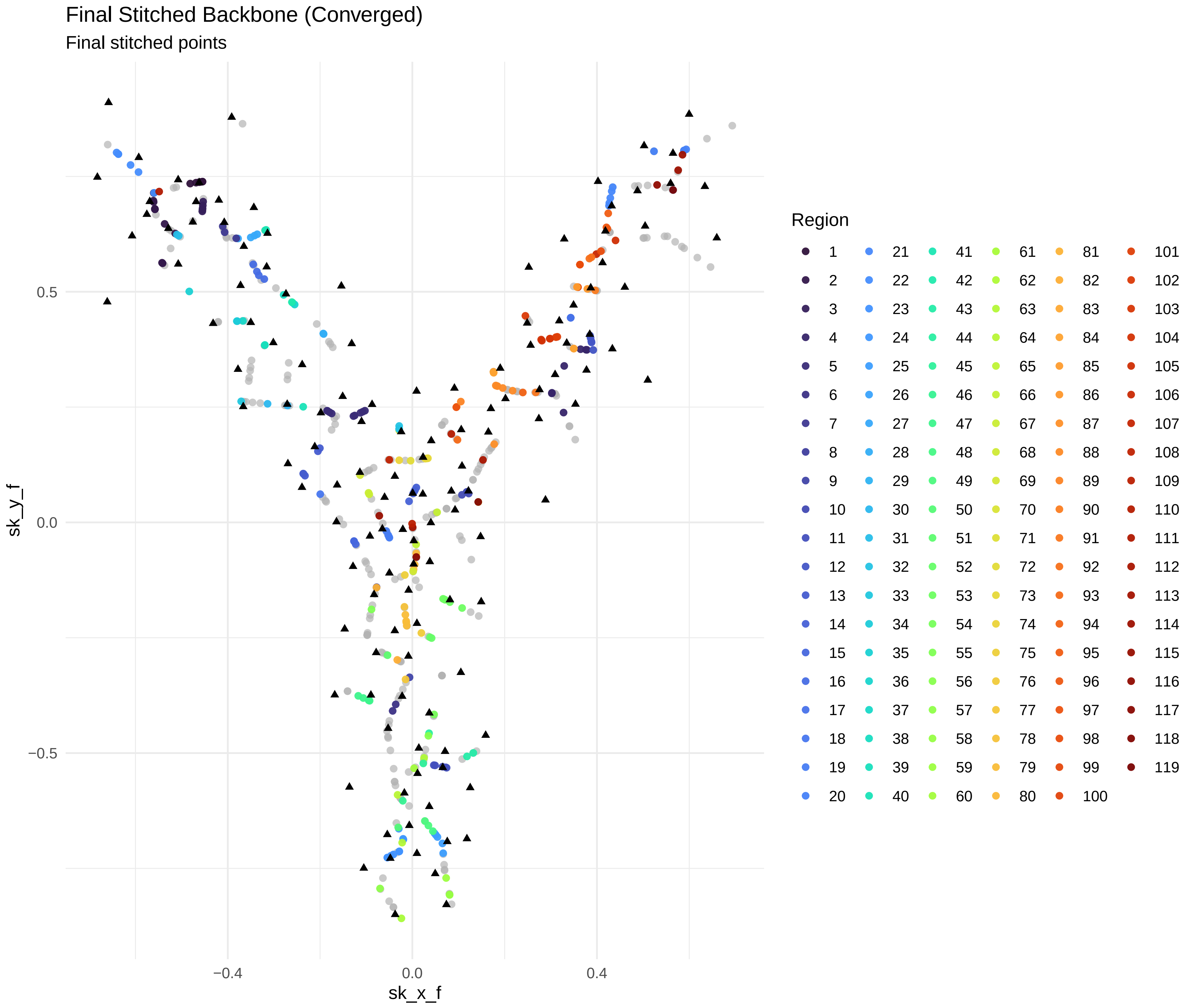
Same anchors as plot 4: spheres first, then their SVD (PC1) in red




Intersection Regions (2+ anchors)
Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.
1.0
0.5
0.0
−0.5
−1.0
y
−0.5
0.0
0.5
x
Region
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80
81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119
s: k_min = 5, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01



Stitched Backbone (Iteration 1)
Orange path: stitching; Triangles: original anchors
y
x
1.0
0.5
0.0
−0.5
−1.0
−0.4
0.0
0.4
Region
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80
81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119


Final Stitched Backbone (Converged)
Final stitched points
sk_y_f
0.5
0.0
−0.5
−0.4
0.0
0.4
sk_x_f
Region
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
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17
18
19
20
21
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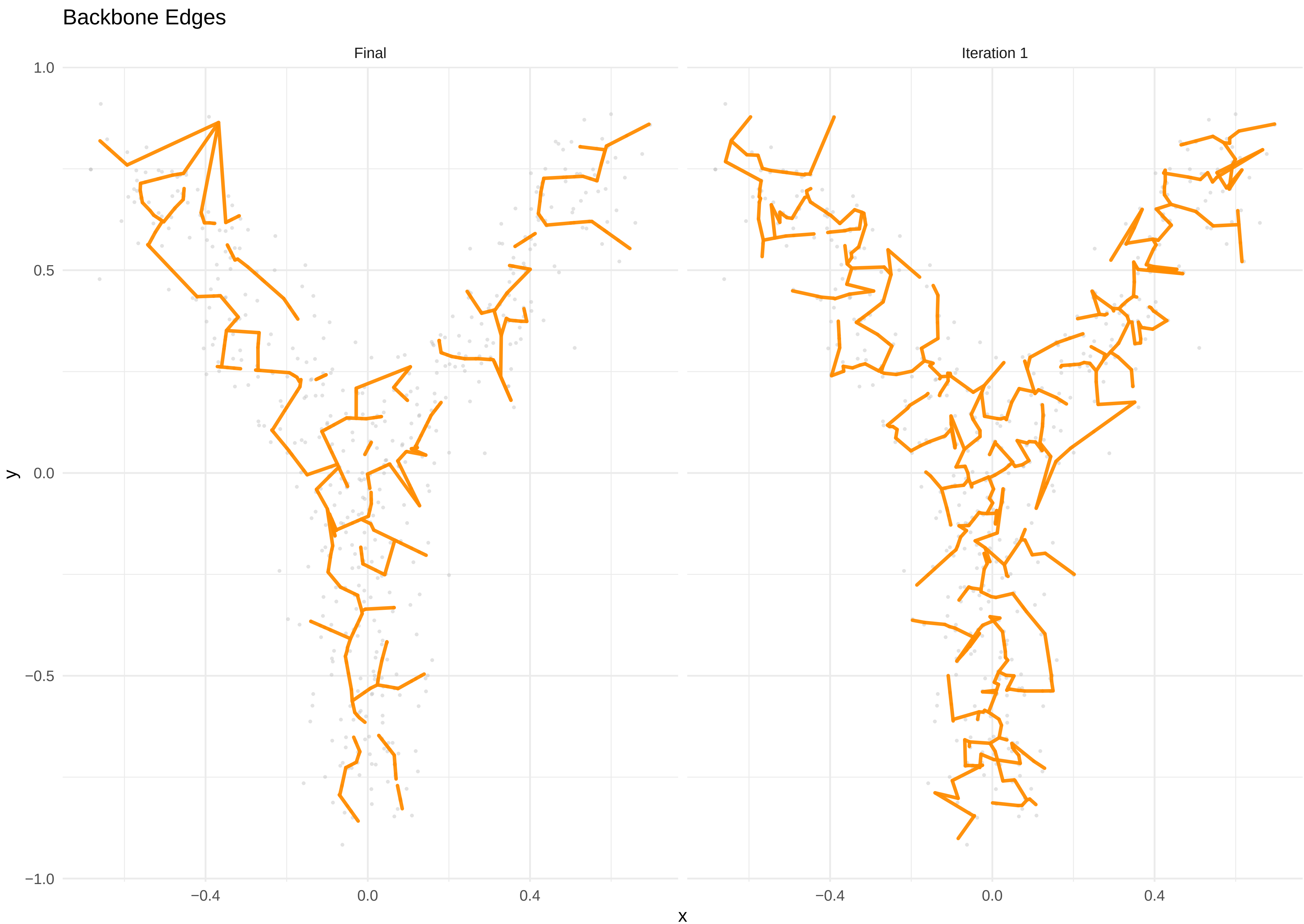
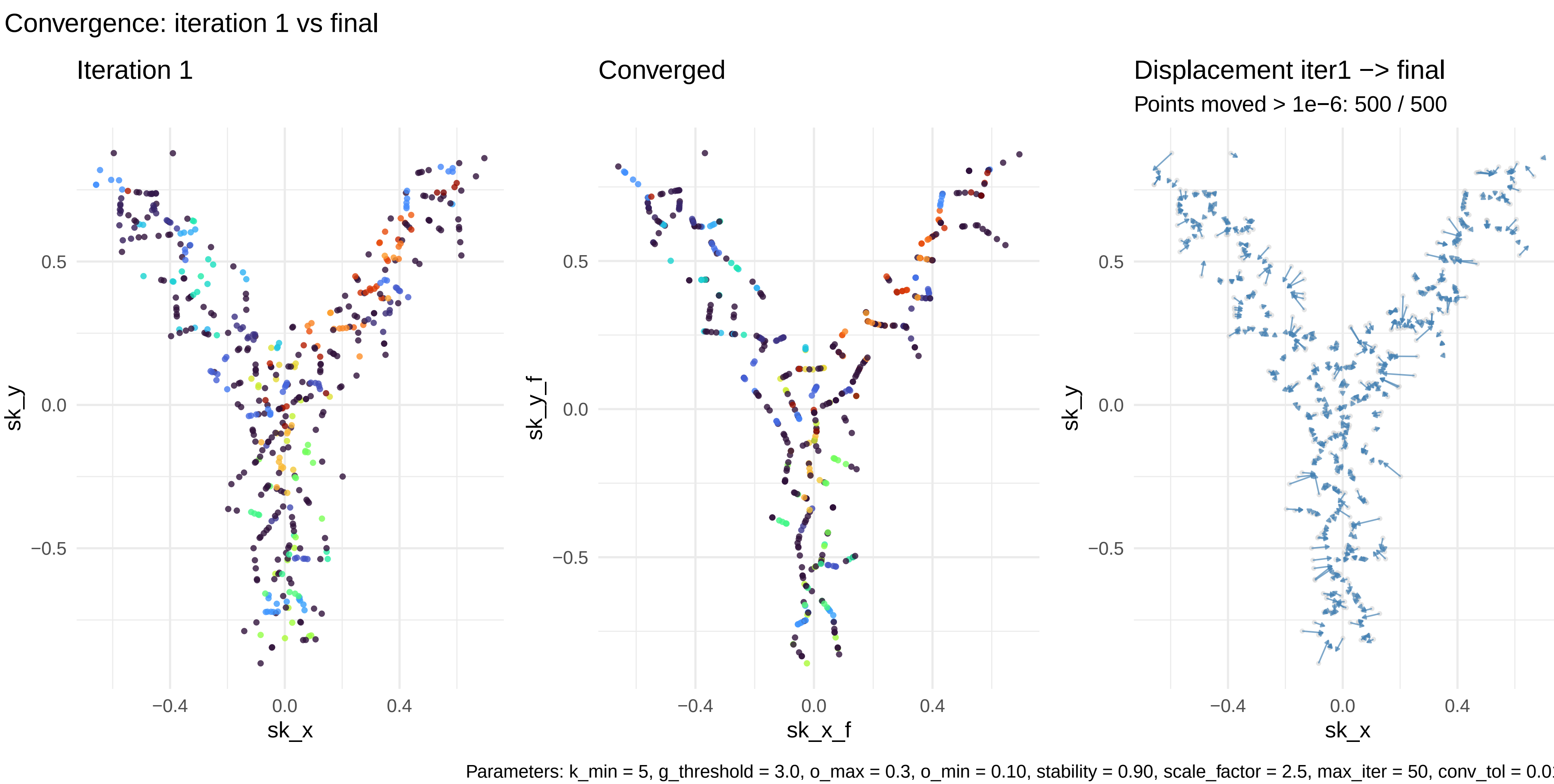

Backbone Edges
Final
Iteration 1
1.0
0.5
0.0
−0.5
−1.0
y
−0.4
0.0
0.4
x



Convergence: iteration 1 vs final
Iteration 1
Converged
Displacement iter1 −> final
Points moved > 1e−6: 500 / 500
sk_y
sk_x
sk_y_f
sk_x_f
0.5
0.0
−0.5
−0.4
0.4
Parameters: k_min = 5, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01